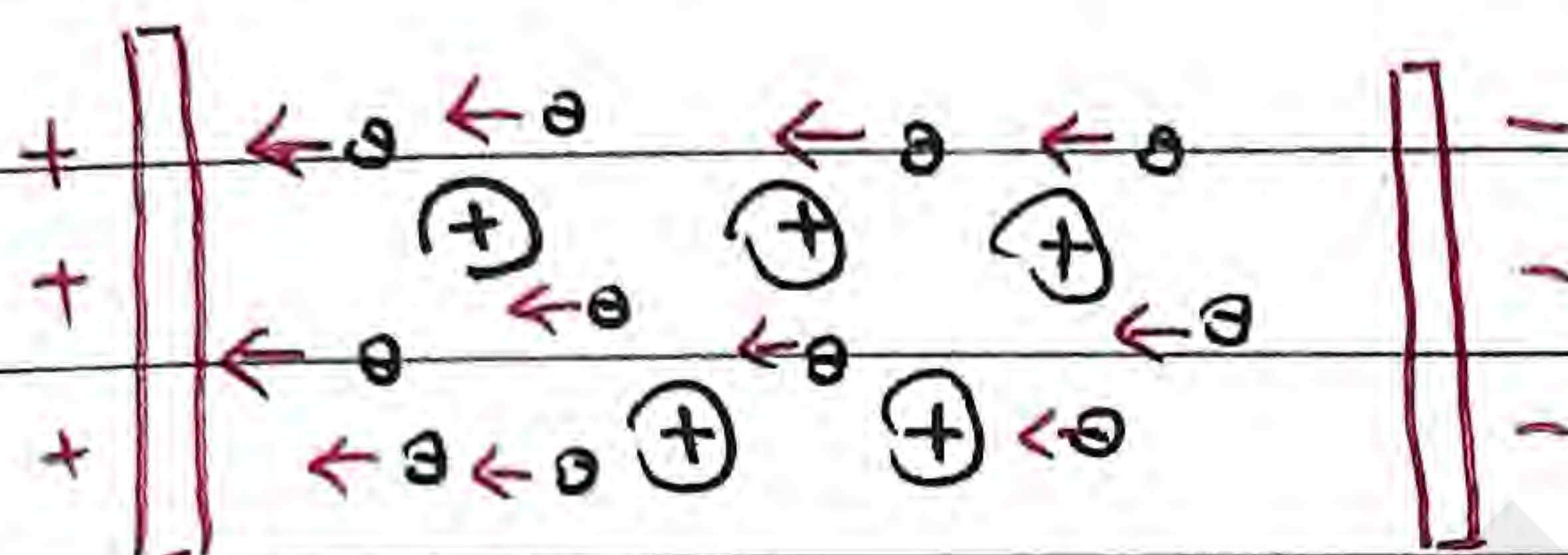


# Electric current

When do charges move?

- When there's a force exerted on the charge
- In other words, when there's an electric field

Let's consider a lattice structure



Let's add  
a uniform electric  
field

Adding a uniform electric field means, probabilistically, most electrons flow in the same direction

- Only electrons move because of lattice structure (more energetically favorable)

\* Electrons do not move linearly due to electron-electron repulsion and electron-nucleus attraction. It's more zig-zag than linear.

The direction of electric current is defined as that opposite to the motion of electrons

$\leftarrow e \leftarrow e$

$\xrightarrow{\quad}$

$I = \text{electric current}$

Electrons also move really slowly ( $10^{-2} \text{ m s}^{-1}$  magnitude)

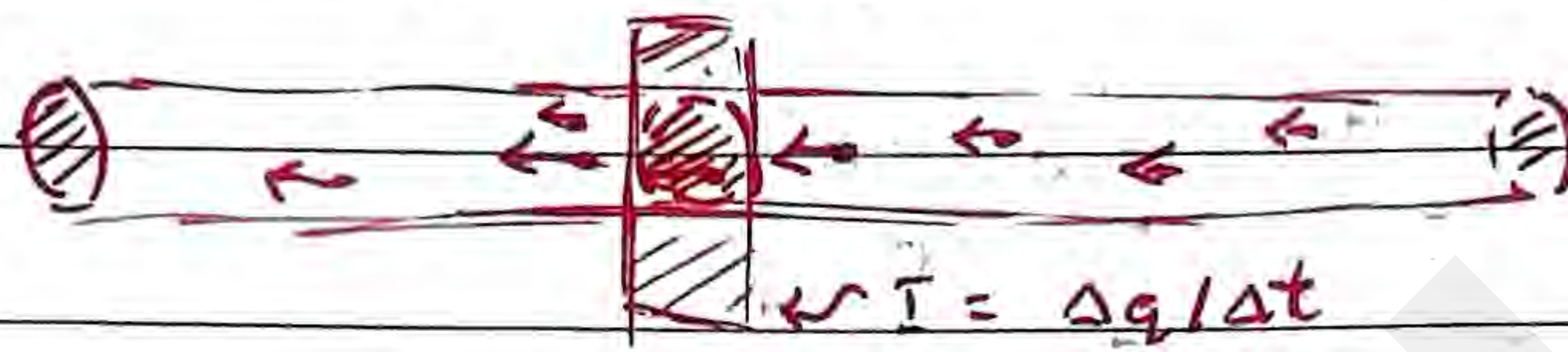
• However, electricity is instant due to fields, which instantly exert force on  $e^-$



## Electric current

Defined as change in charges per reference area in a given time unit (second)

$$I = \frac{\Delta q}{\Delta t}$$

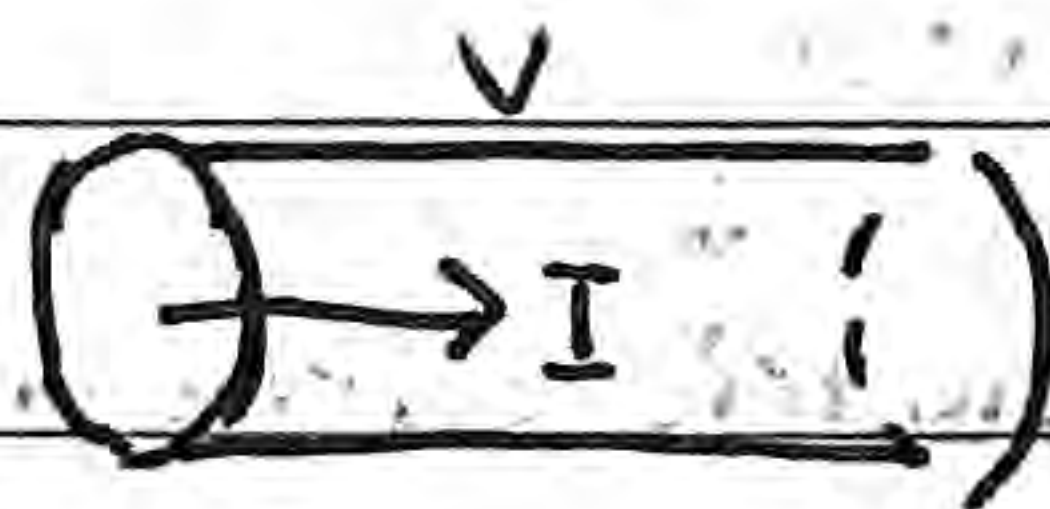


where units are  $C \cdot s^{-1} = A = \text{Ampere}$

## Electric resistance

While friction certainly does not exist in a subatomic level, there's repulsive and resistive forces against electrons.

This is known as resistance, defined as the potential difference over the current in a given conductor



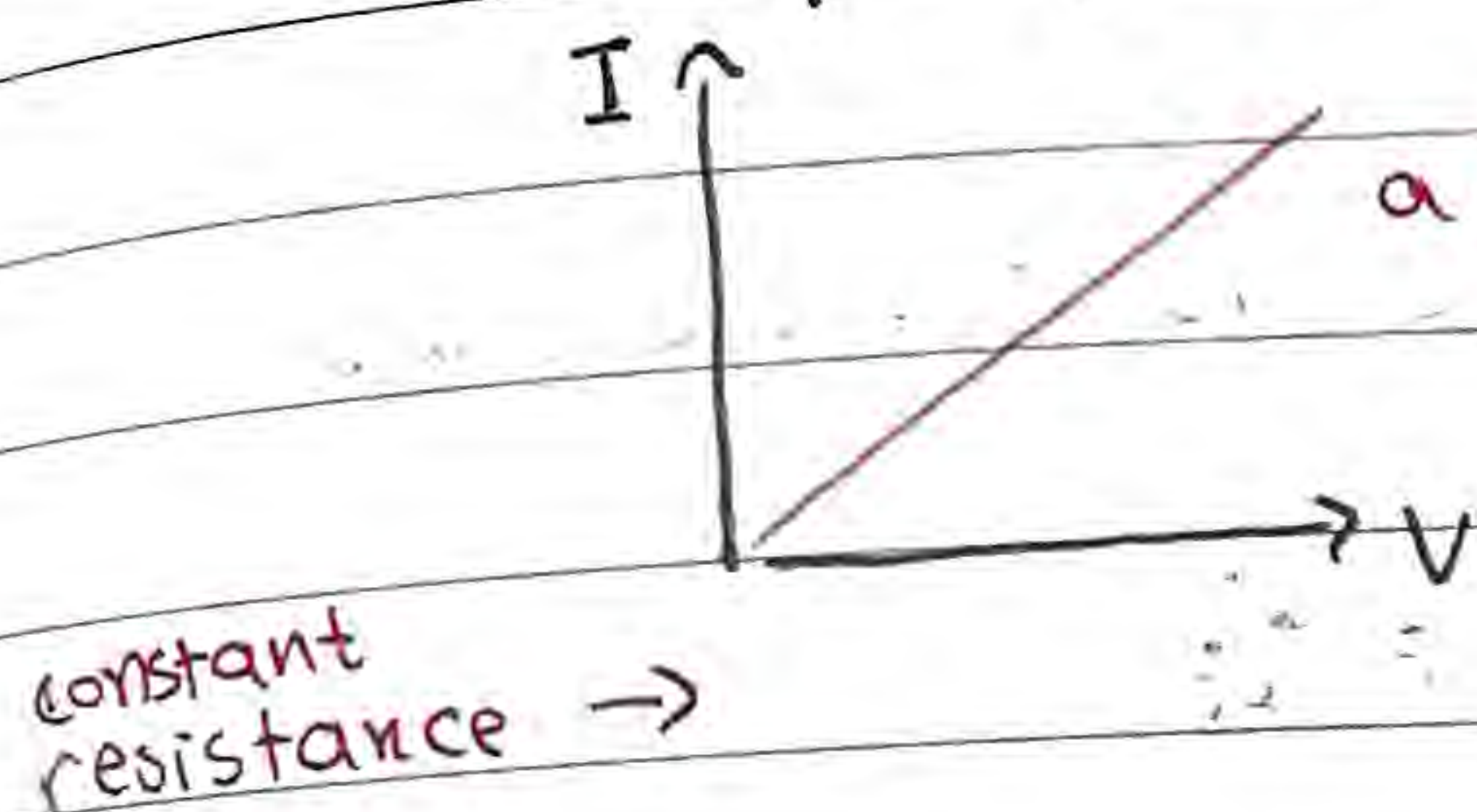
$$R = \frac{V}{I} [V \cdot A^{-1} = \Omega = \text{Ohm}]$$



## Electric resistance

When you measure current per different values of voltage, you can calculate the resistance of the material upon a change in voltage (applied)

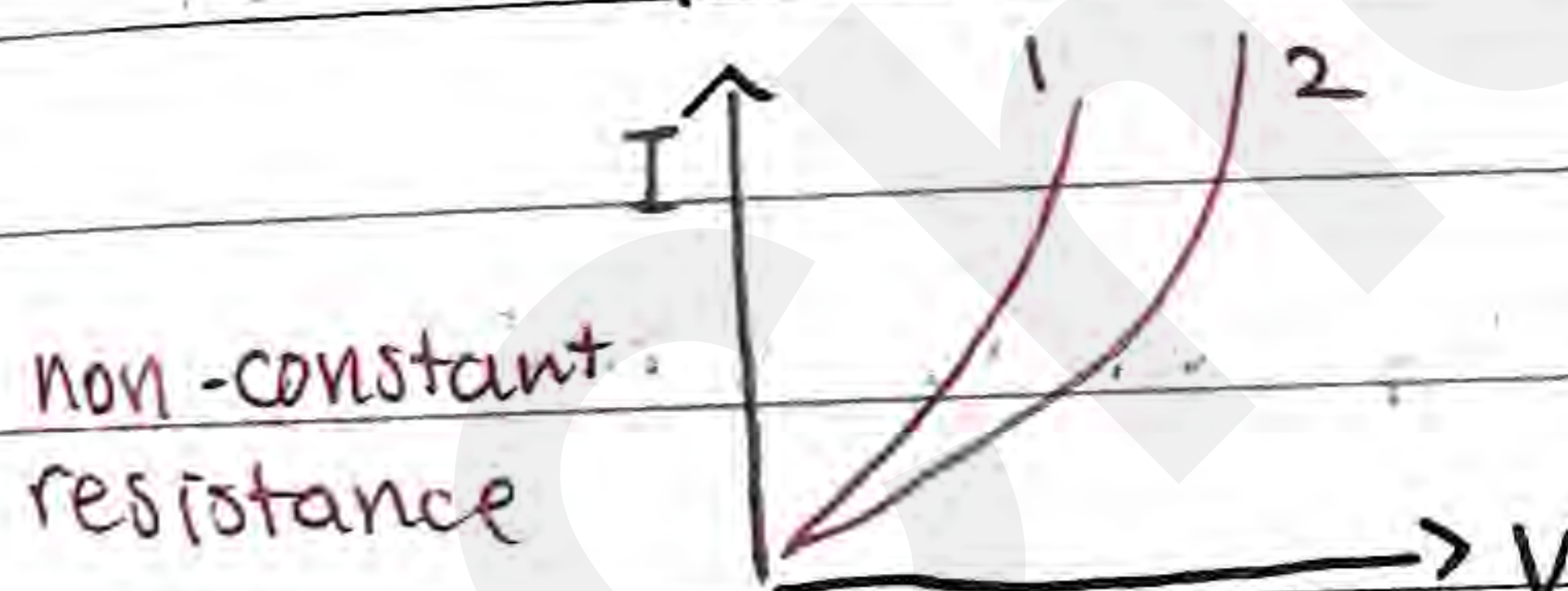
- Graphing them out:



a perfectly linear relation

This is an ohmic material  
(e.g. resistors)

However... some materials do not have a perfectly linear relation



This is a non-ohmic material  
• All natural materials are non-ohmic

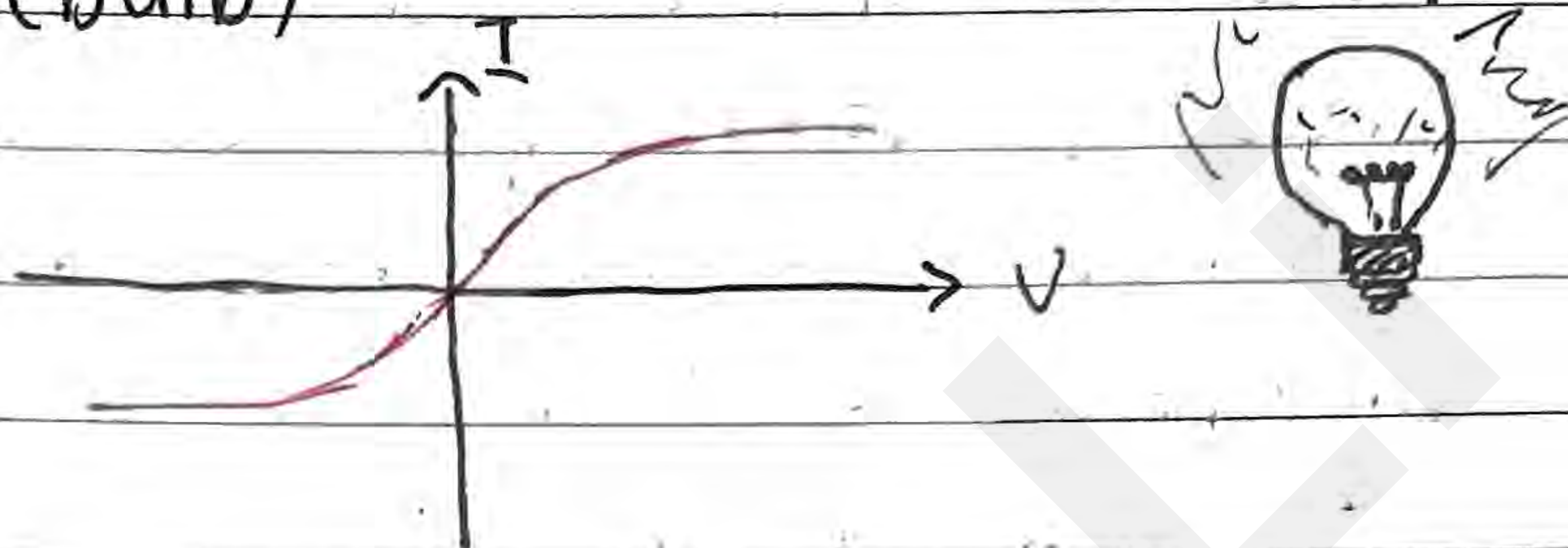
In circuits, everything is composed of these five components:

- |                     |  |
|---------------------|--|
| 1. Resistor         |  |
| 2. Diode            |  |
| 3. Capacitor        |  |
| 4. coil / Inductors |  |
| 5. Transistors      |  |

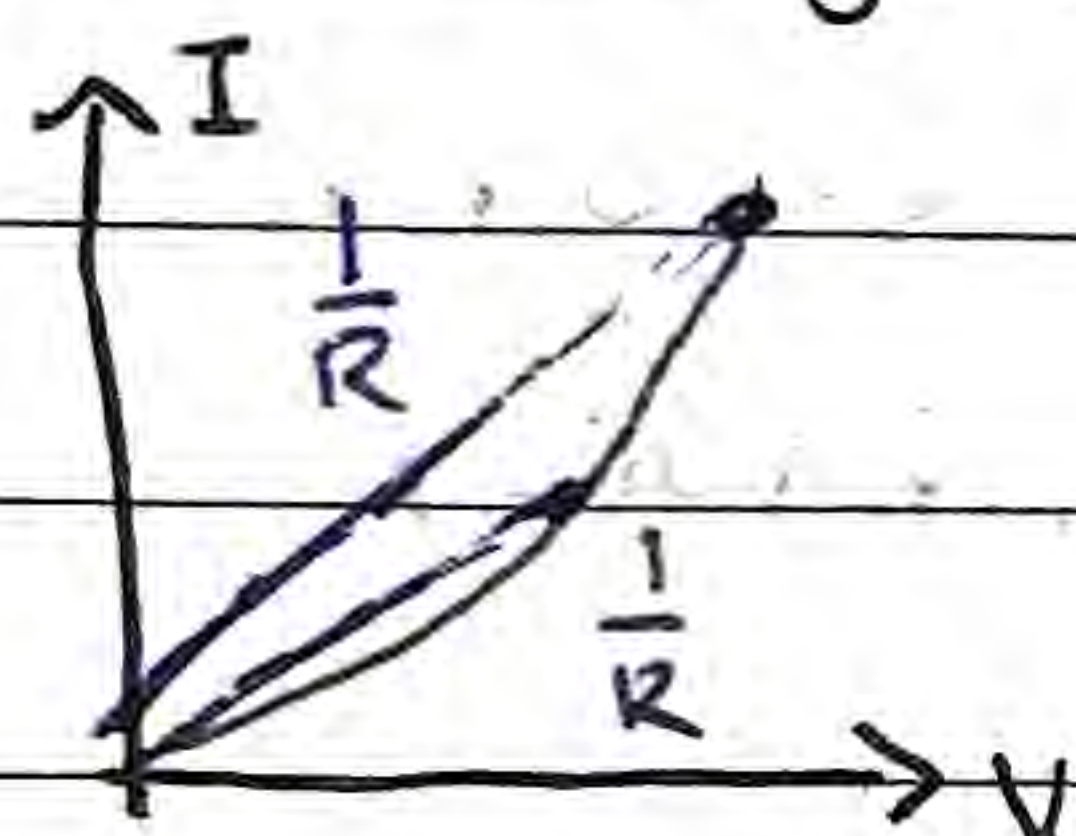


## I-V graph

Filament (bulb)



In a I-V graph, resistance is NOT the slope



$$R = \frac{V}{I} \neq \frac{\Delta V}{\Delta I}$$

it's the reciprocal of the average rate of change from  $V=0$ . (not the instantaneous!)

Resistance is affected by various factors

The following formula illustrates this relationship

↑  
for a conductor

$$R = \rho \cdot \frac{L}{A}$$

where ...

$R$  - Resistance of object

depends on temperature →

$\rho$  - Resistivity of material ( $\Omega \cdot m$ )

$L$  - Length of object

$A$  - Cross-sectional area

|               |                |
|---------------|----------------|
| $\Rightarrow$ | $R \downarrow$ |
| $\Rightarrow$ | $R \uparrow$   |
| $\Rightarrow$ | $R \uparrow$   |
| $\Rightarrow$ | $R \downarrow$ |
| $\Rightarrow$ | $R \uparrow$   |



## Electrical Conductivity

$$\text{Electrical Conductivity} = \frac{1}{R}$$

more resistive, less conductive

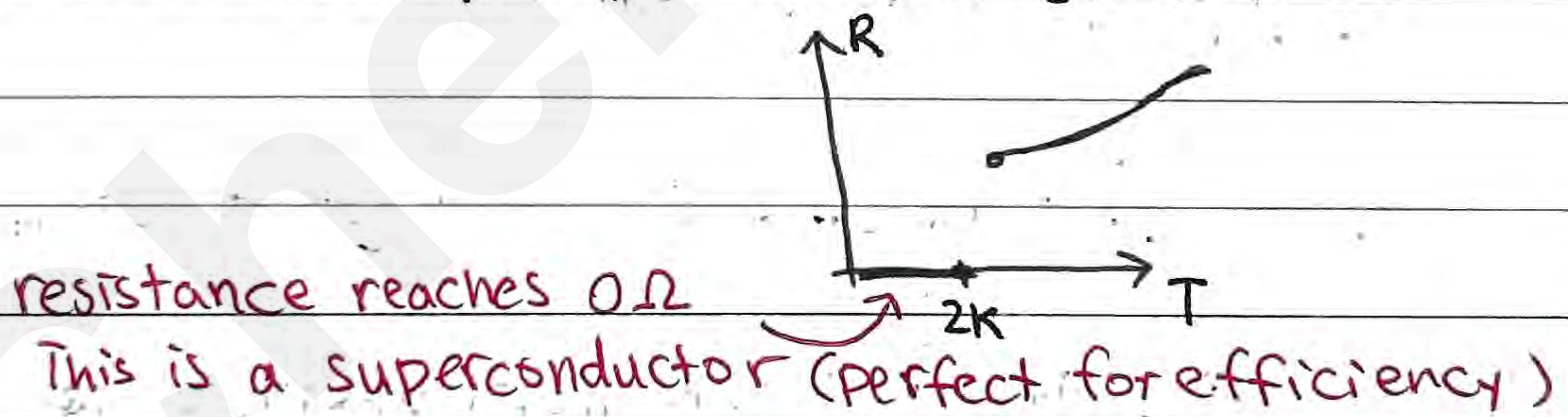
less resistive, more conductive

## Temperature and resistance

As temperature increases, resistance increases.

Lower temperature, lower resistance

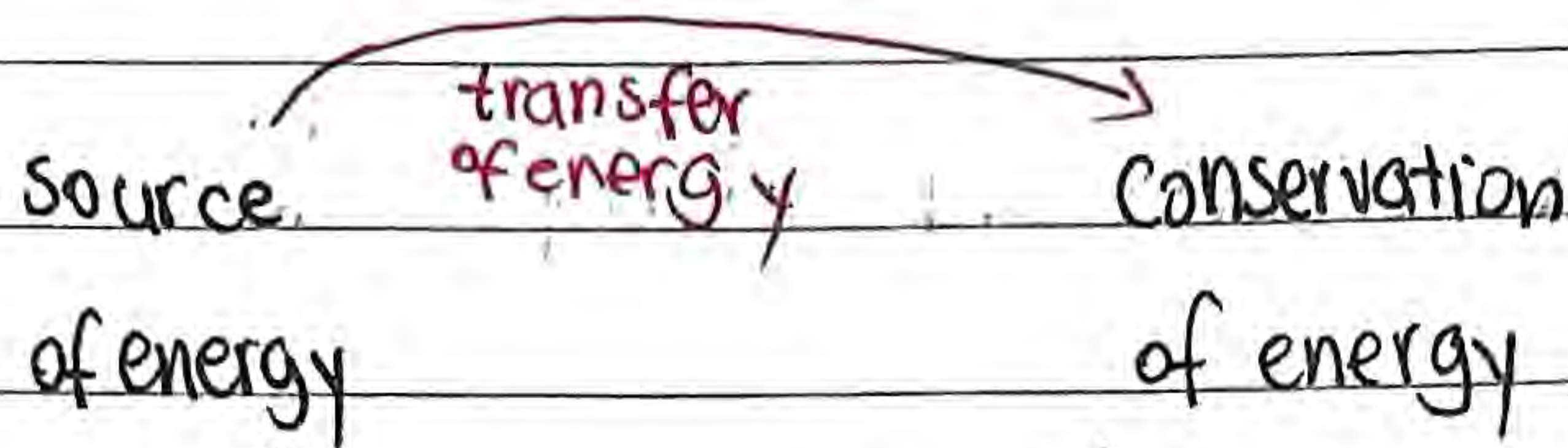
but at some point, resistance jumps to 0





# Electricity

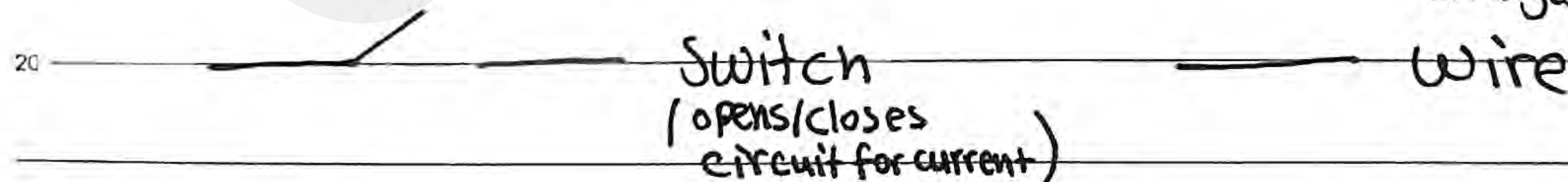
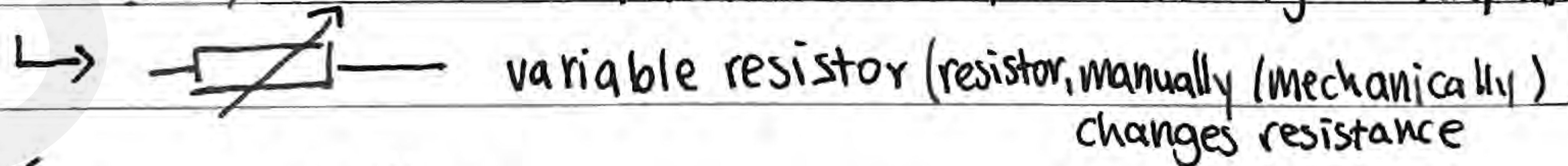
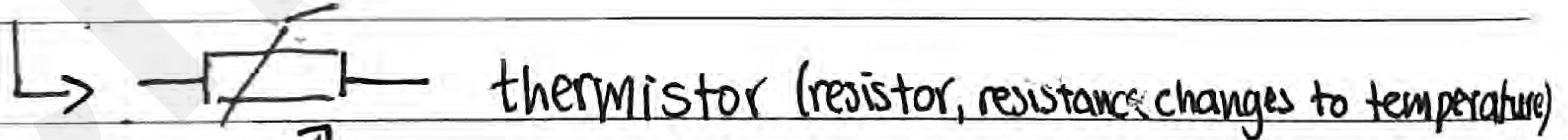
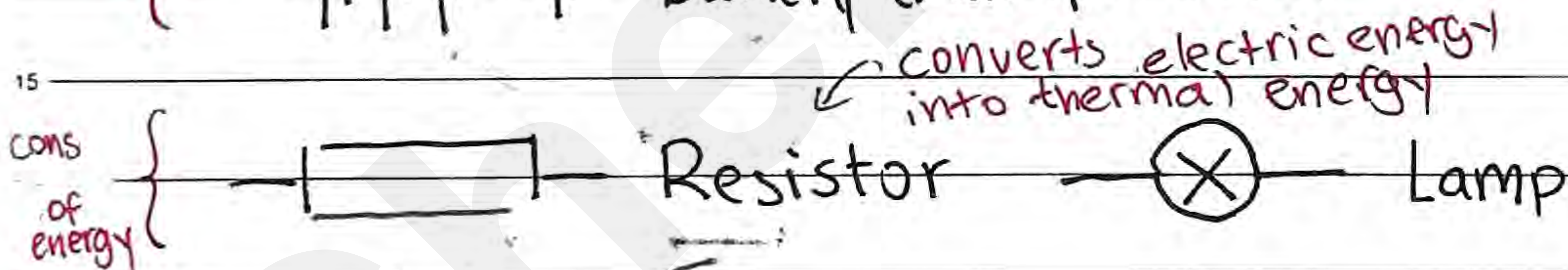
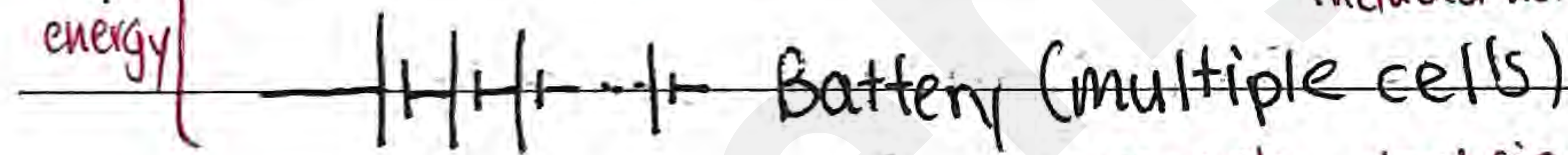
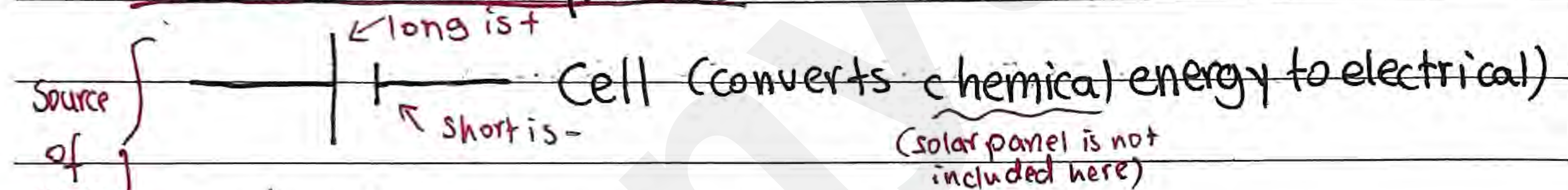
Electric circuit - To transfer energy from source to consumer



but you also need to close the loop!

In order to create a potential difference, you need a "pump" or a battery

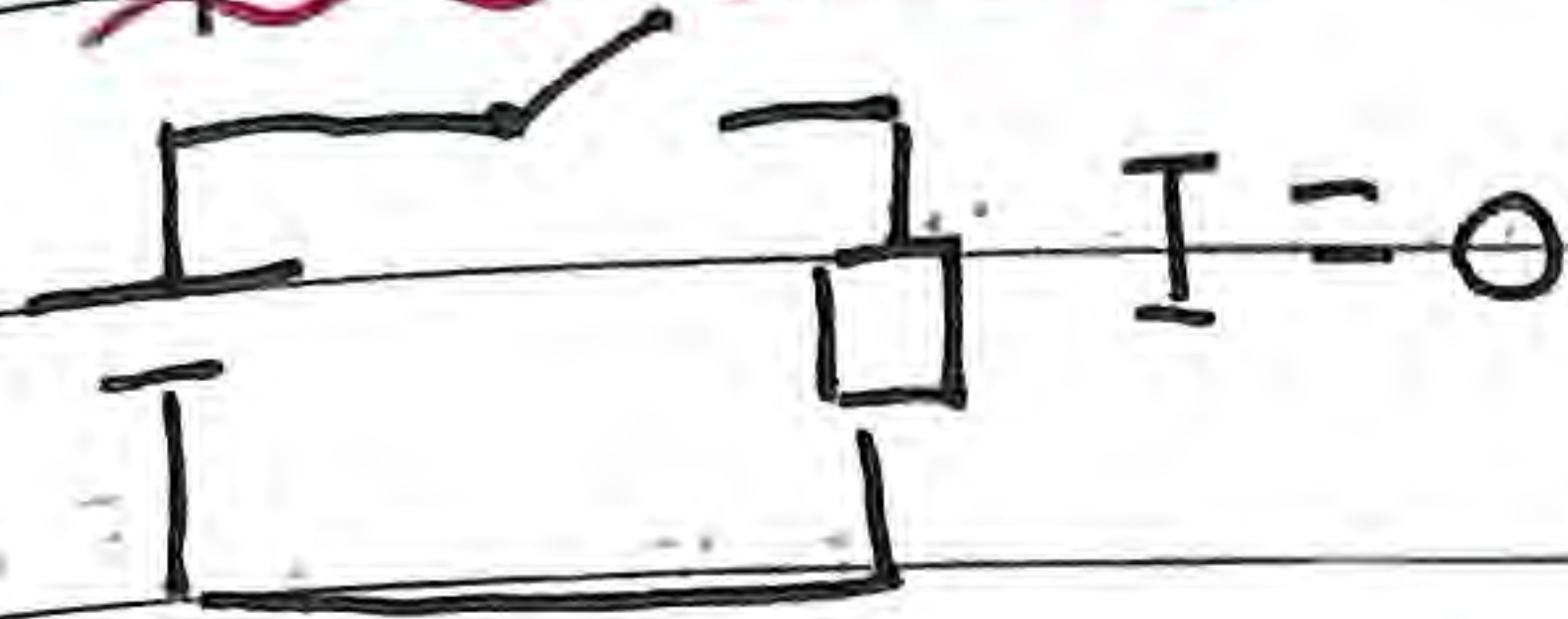
## Electric Circuit Symbols





## Basic Electric circuit

Open circuit

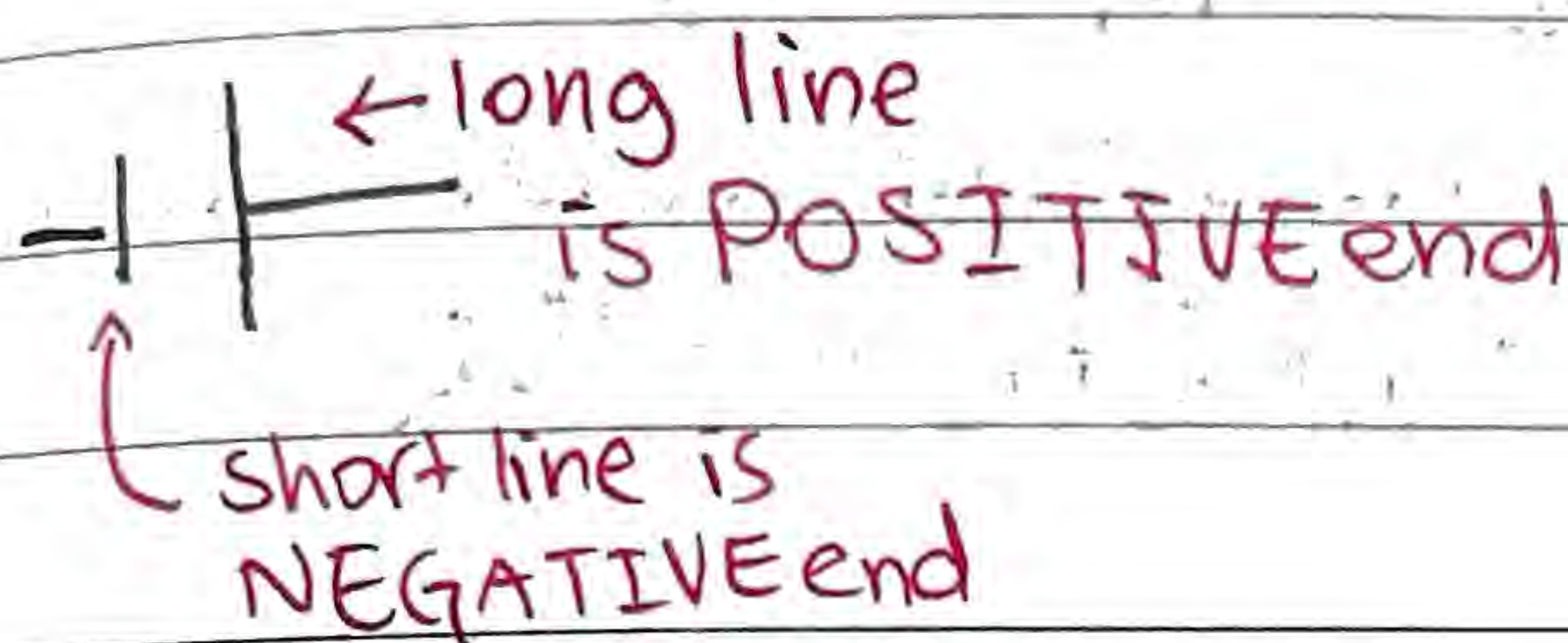


Closed circuit



opposes  
flow of electron

Current is positive to  
negative end



Ohm's law:

$$R = \frac{V}{I} = \frac{\text{Potential Difference}}{\text{current}}$$

E.g.  Then current =  $\frac{3}{2} \text{ A} = 1.5 \text{ A}$

However, circuits are not that simple...

Resistors in series

terminals  
do not  
meet/  
not the  
same



The current through each resistor

Equivalence  
Resistance =  $\sum R_i$

is the same

=  $R_1 + R_2 + R_3 + \dots$

As such, according to Ohm's law,

Total  
Voltage =  $\sum V_i$

the Potential Difference on each resistor  
is not the same

=  $V_1 + V_2 + V_3 + \dots$



## Resistors in parallel



All three terminals of the resistors meet / are of the same point

• The potential difference on each resistor is the same

• The Equivalent Resistance is calculated as follows:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n} = \sum \frac{1}{R_i}$$

• According to Ohm's law, the current on each resistor is not the same:  $I = I_1 + I_2 + I_3 \dots = \sum I_i$

You can think of it like this

In series



- Same "wire", same current
- Different terminals, different voltages

In parallel



- Different "wire", different currents
- Same terminal, same voltage

## Electric Power

- You can calculate the energy passing through any component in a given amount of time

⇒ Depending on whether it is source or consumer of energy, it will be power released or absorbed respectively

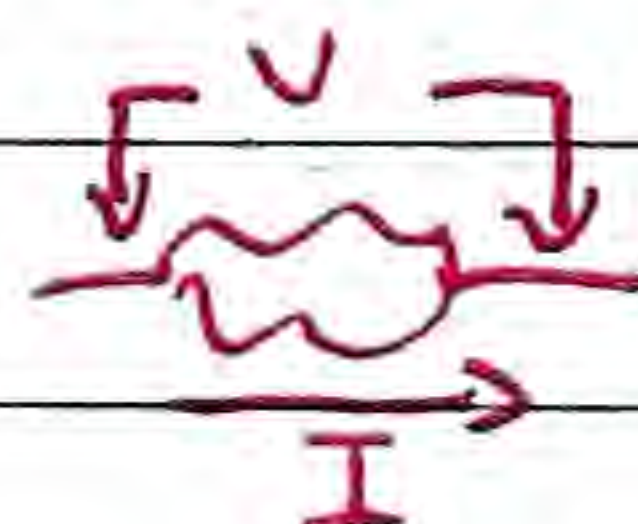
total power  
=  $\sum P_i$   
(for both series, parallel)

$$P = I^2 \cdot R = V \cdot I$$

where...

$$\hookrightarrow V = I \cdot R$$

$$P = \frac{V^2}{R}$$



P - Electric power of component

R - Resistance of component

I - Current passing through

V - Potential difference of terminals

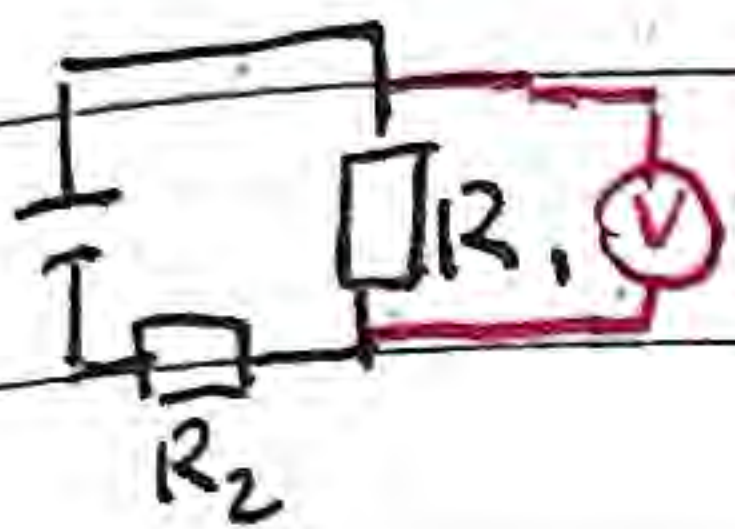


Ideal Ammeter - An instrument measuring the current in a series such that the resistance is negligible



↳ to allow for all current to pass through ammeter

Ideal Voltmeter - An instrument measuring the potential difference in parallel such that the resistance is infinitely large

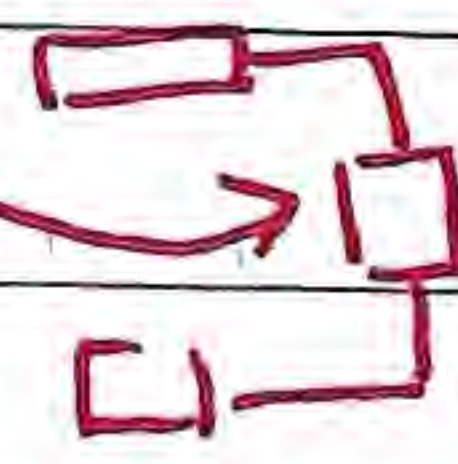
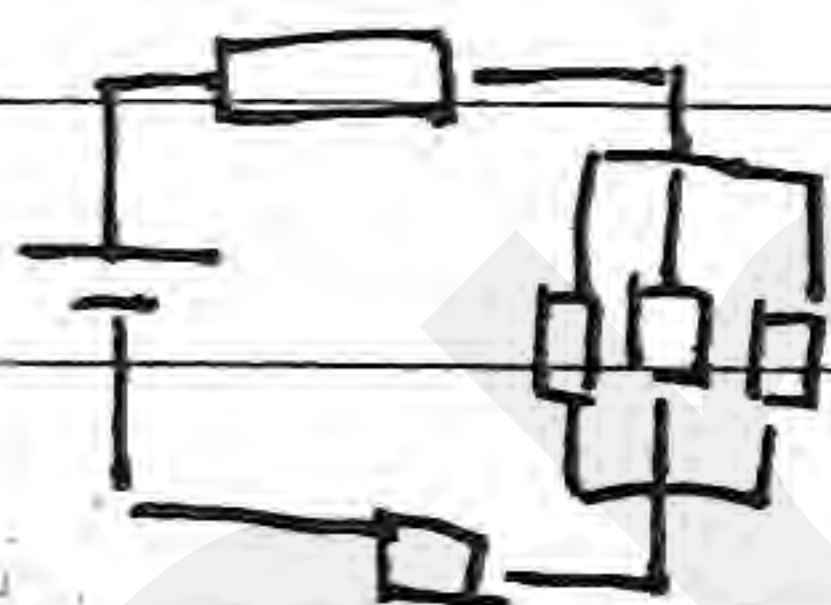


↳ to prevent current from escaping to voltmeter

Complex circuits often consist of in series and in parallel components

↳ You can group these different components to do mini calculations

(go from innermost circuit, the one you're 100% certain with, of the same type)

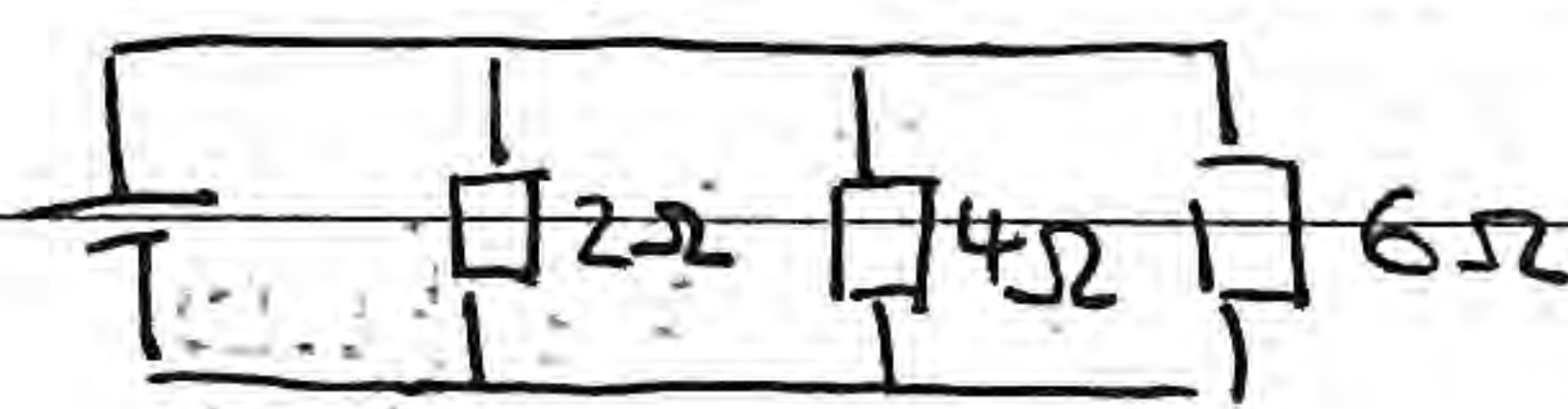
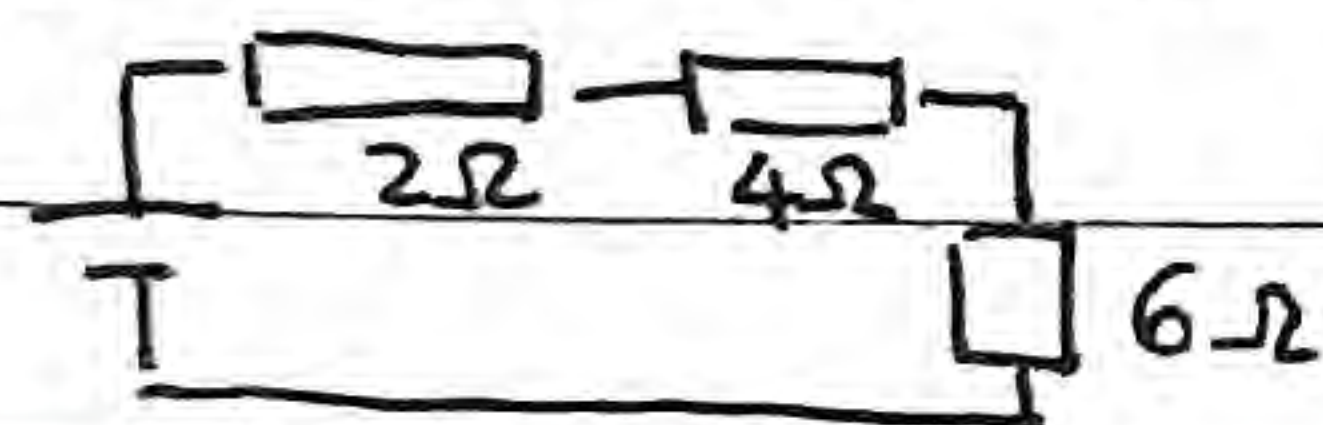


to then calculate  $R_{eq}$  for overall circuit

After you find current of entire circuit, you go inwards one-by-one

Analyzing parallel and series circuits

E.g. For 24V circuit with resistors of  $2\Omega$ ,  $4\Omega$ ,  $6\Omega$



$R_{eq}$  lower for parallel, greater current, greater power (useful for high power)

$R_{eq}$  higher for series, lower current, lower power (useful for low power)



## Types of resistors

> Thermistor - Temperature dependent resistance  

 • As temperature increases, resistance decreases.

> Light dependent resistor - Intensity of light dependent resistance  

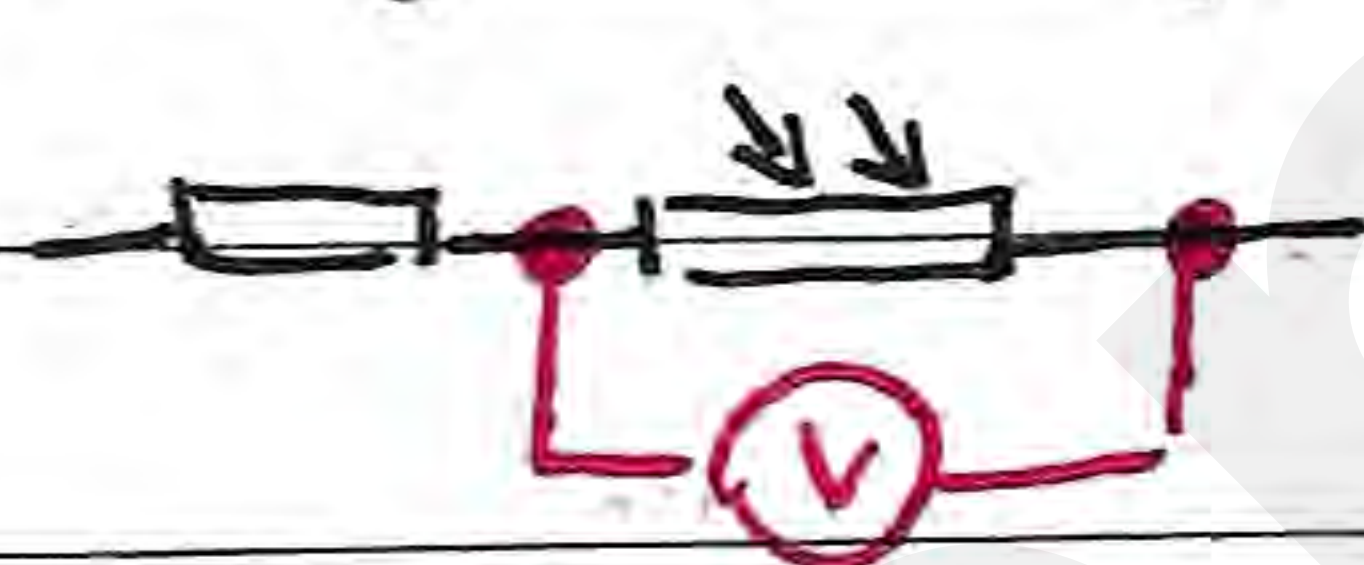
 (LDR)

• As intensity of light increases, resistance decreases

> Potentiometer - Length dependent resistance  

 • As length of resistor increases, resistance increases

E.g. Given constant current, what happens to measured voltage as light intensity increases



$I \uparrow$  thus  $R \downarrow$

Given  $V = I \cdot R$

$V \propto R \rightarrow R \downarrow$  so  $V \downarrow$

## Voltage, current, resistance

• As we know, voltage stays constant in a circuit connected in parallel, current stays constant in a circuit connected in series.

• As such, we can also get the following formula...

$$V = IR$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{I_1 \cdot R_1}{I_2 \cdot R_2}$$

For constant voltage (parallel)

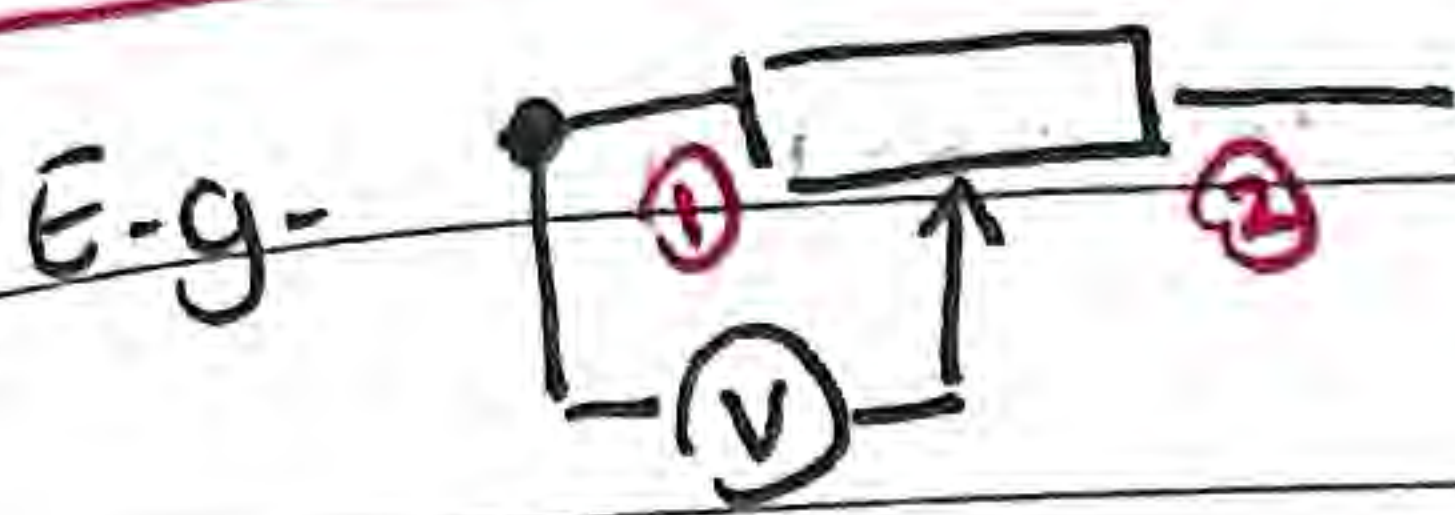
For constant current (series)

$$\frac{I_2}{I_1} = \frac{R_1}{R_2}$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

Can be useful when one resistance is equivalence resistance



Potentiometer

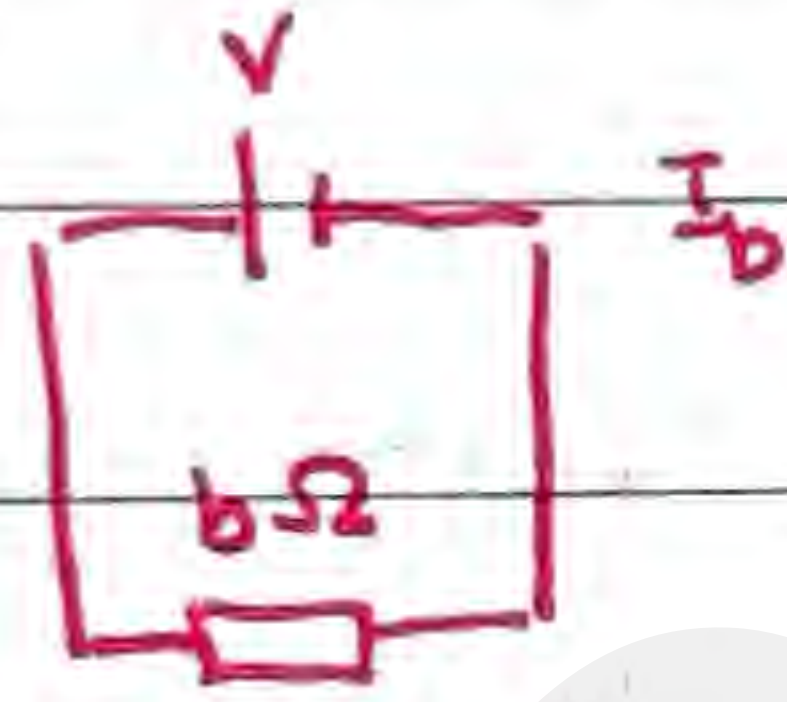
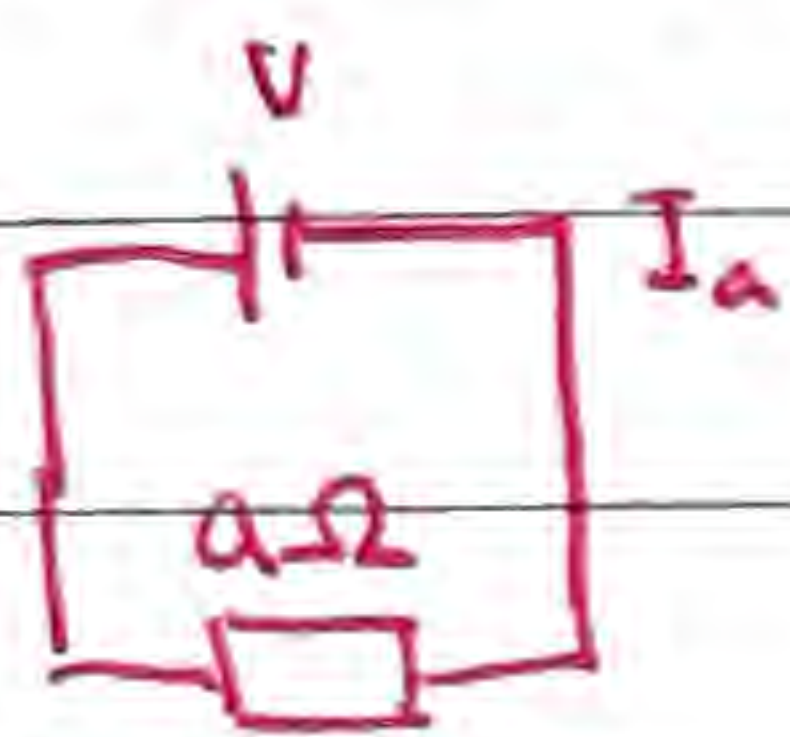
the labelled resistance is the maximum resistance

At ①, voltmeter does not measure a potential difference,  $V = 0$

At ②, voltmeter measures maximum potential difference

Voltage, current, resistance

When resistance changes in a circuit, so will the current (assuming voltage is constant due to cell)



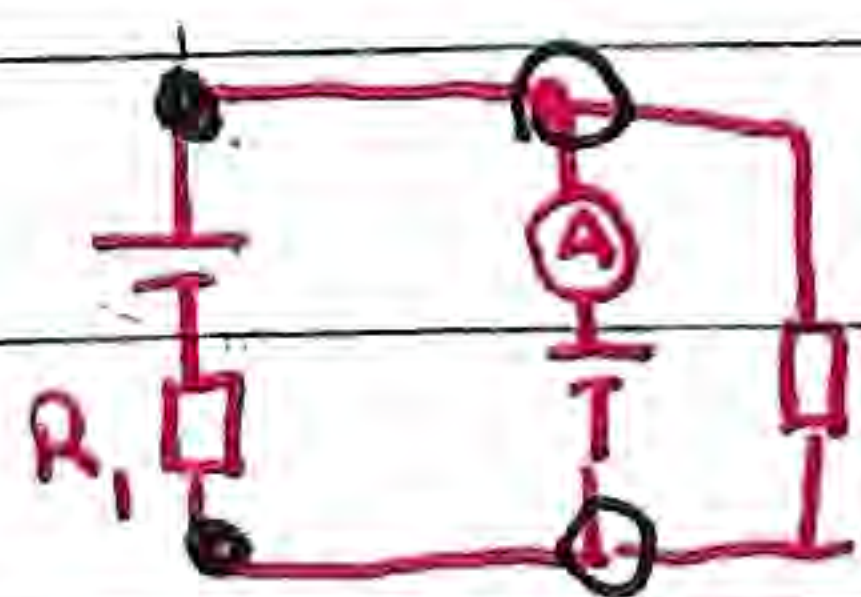
if  $b > a$ , then  $I_b < I_a$

So, when a problem requires a change in resistance, make sure to recalculate the current

↳ To find voltage, power, etc.

Power consumed = Power delivered

$$\sum P_i \text{ of individual circuits} = \text{Power of system} = I \cdot V$$

Multiple cell battery

two terminals are connected in parallel,  $V$  is same

You know  $V$  of  $\text{cell}$  is  $6V$ , but not

↳  $R_1$  must have  $0A$  passing through ← this also  $6V$  ← ∴ (since there's resistance)

or else would decrease (taken out)

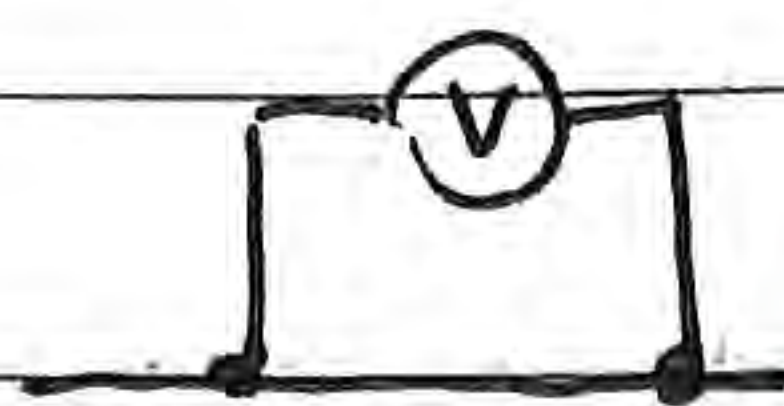


## Complicated electric circuits

• Remember an ideal voltmeter has  $\infty$  resistance

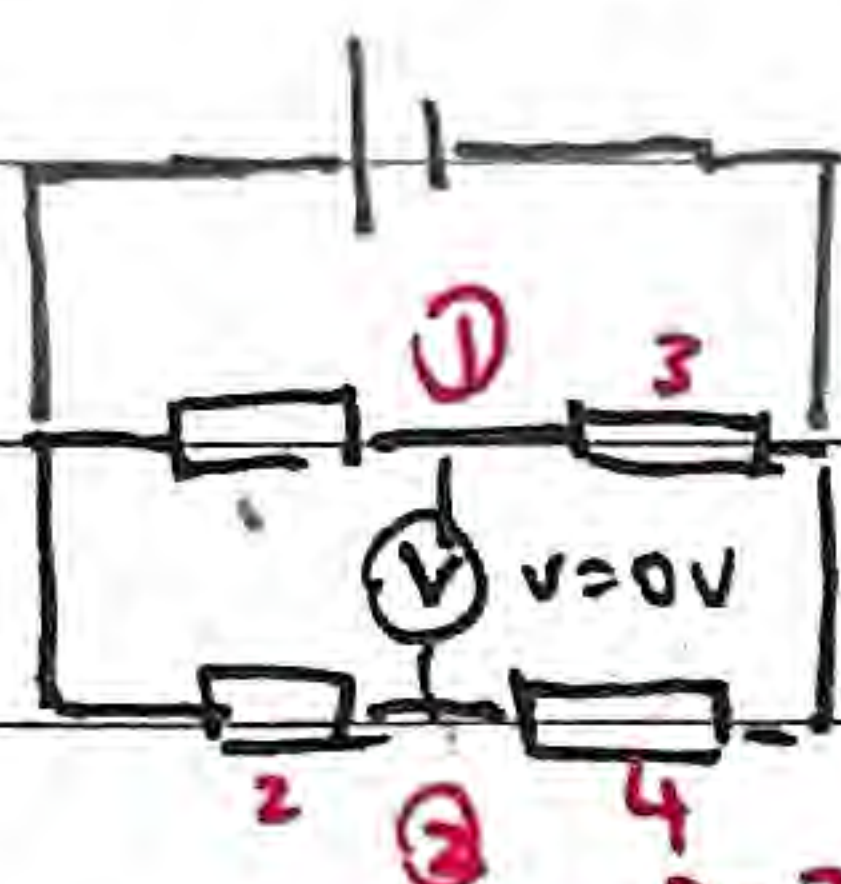
• And, when...

$$V = 0 \text{ V}$$



is the same as no resistor in between

then... E.g.



$V=0$  so energy from cell to ① is same as to ③

$R=?$  thus  $V_1 = V_2$ ,  $V_3 = V_4$

but also, voltmeter allows no current to pass through, so  $I_3 = I_1$  (no splitting),  $I_4 = I_2$

thus

$$\frac{V_1}{V_2} = \frac{V_3}{V_4} \Rightarrow \frac{I_1 R_1}{I_2 R_2} = \frac{I_3 R_3}{I_4 R_4}$$



## Electromotive Force (Emf, $\mathcal{E}$ )

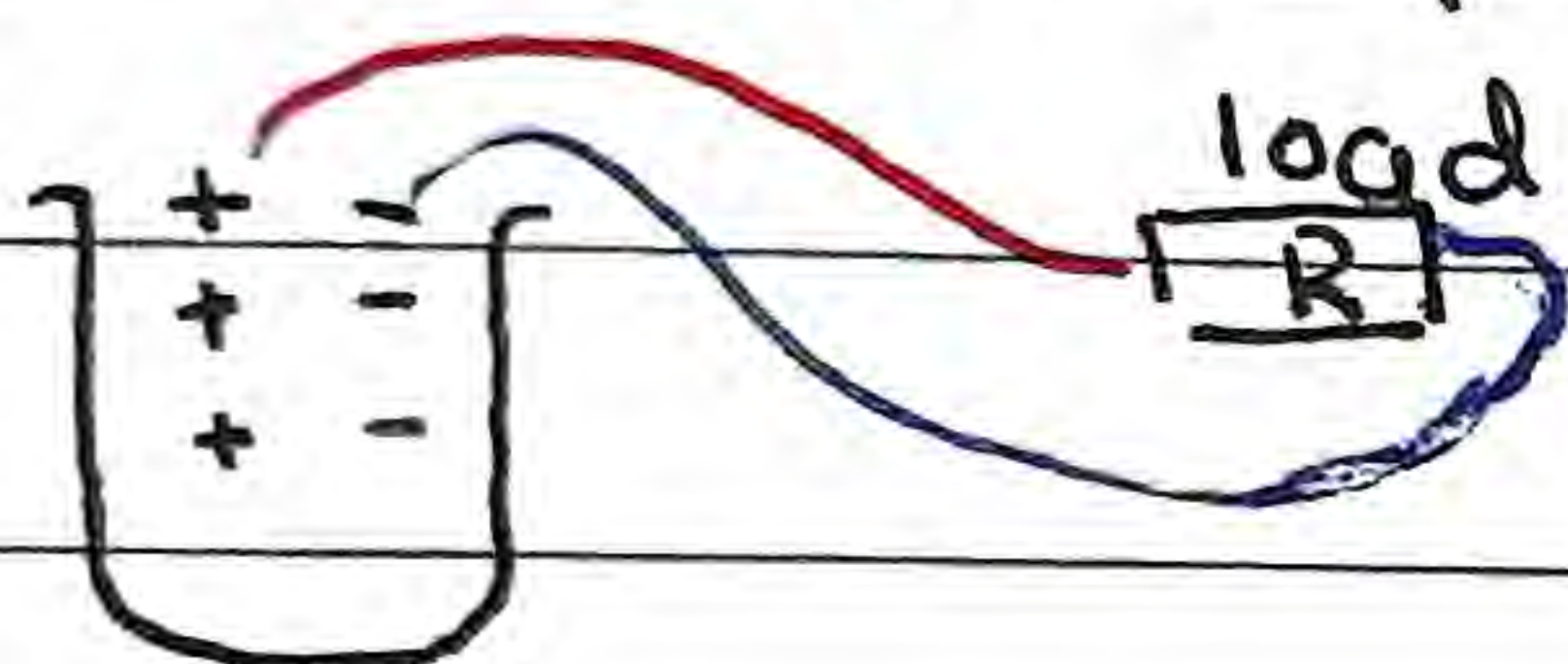
→ Not force, it's energy

The work done per unit charge in moving a charge from one terminal of the battery to the other.

↳ the units are volts, but it's not the same as electrical potential difference.

## Internal Resistance

Recall a battery...



since charges must travel through a medium (non-vacuum)

Thus we can think of it like this...



Real Battery

Ideal Battery



$\mathcal{E}$  is the voltage of an ideal battery and the voltage of a real battery when current = 0 (open circuit)

↳ Thus actual voltage of a real battery in a closed circuit will never reach  $\mathcal{E}$  (advertized voltage)



$$\text{current} = \frac{\mathcal{E}}{R+r}$$

$$\mathcal{E} = I(R+r)$$

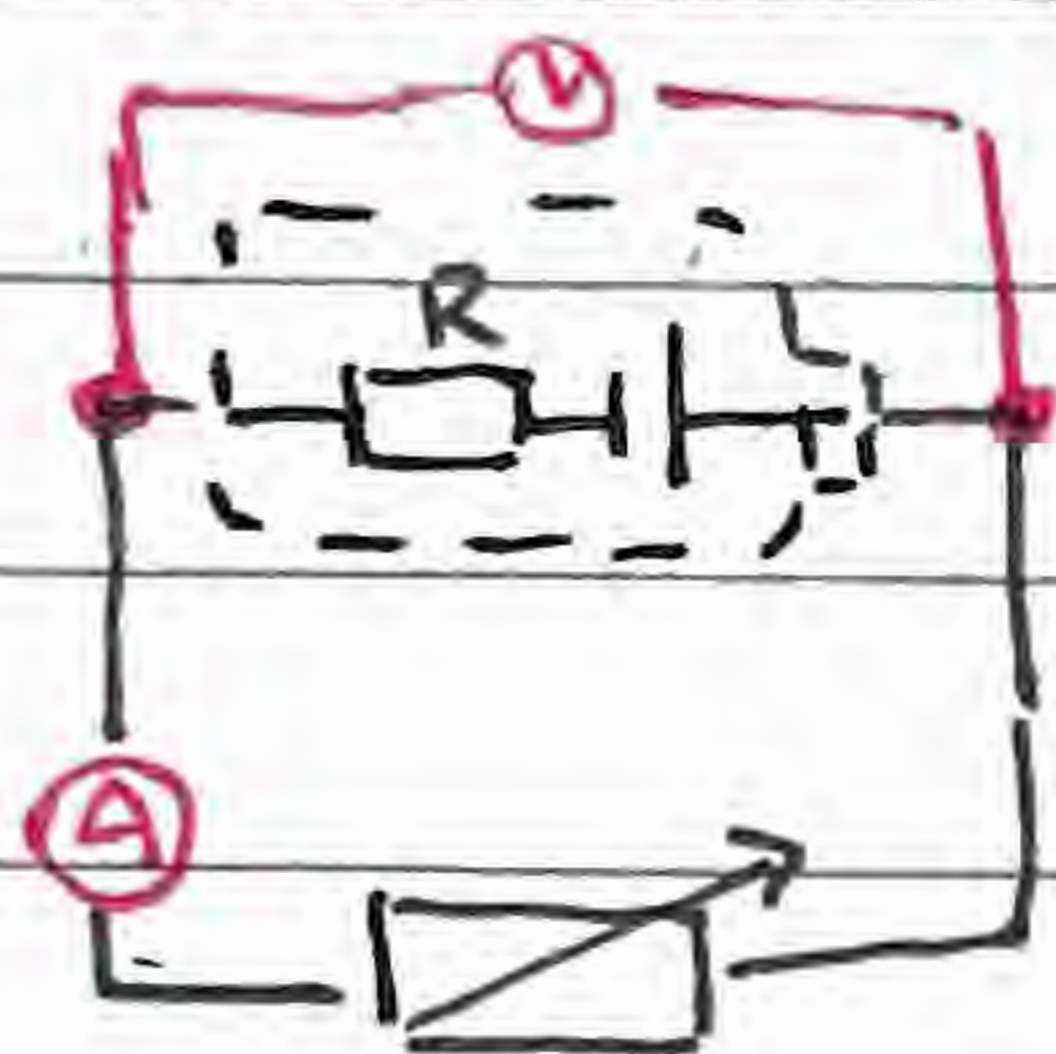
$$\text{Voltage} = \mathcal{E} - Ir$$

↑  
energy taken out



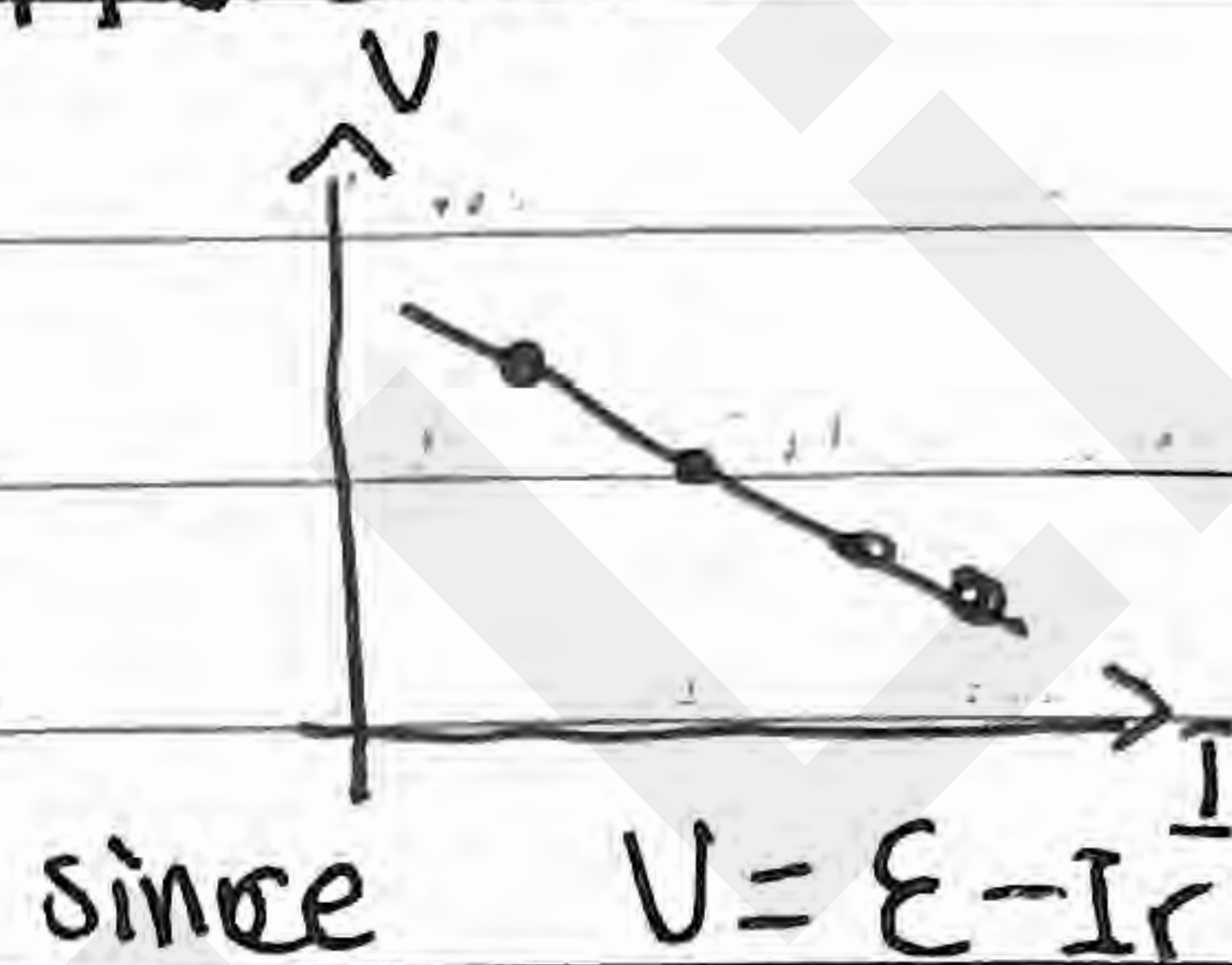
## Measuring internal resistance

- You can measure internal resistance by changing external resistance and measuring current and electrical potential difference



I vs V

|  |  |
|--|--|
|  |  |
|  |  |
|  |  |



so  $\bar{E}$ mf is y-intercept, internal resistance is slope